NOTES FOR MAT1101

OLLE RENÉRIUS REHNQUIST

1. Lecture 1

1.1. **Motivation for Galois Theory.** The first part of the course is Galois theory, for which the original motivation was to understand solvability of polynomials using radicals. Galois had the insight that one should aim to understand the symmetries among the roots.

Example 1.1. Consider the polynomial equation $x^3 - 3x + 1 = 0$ which is irreducible over $\mathbb{Q}$. If $\alpha \in \mathbb{C}$ is a root, then the others are given by $1 - \frac{1}{\alpha}$ and $\frac{1}{1-\alpha}$. This situation is not typical for cubic polynomials. One can then form the subfield $\mathbb{Q}(\alpha) \subseteq \mathbb{C}$ generated by the roots. Since $\alpha^3 = 3\alpha - 1$ we find that

$$\mathbb{Q}(\alpha) = \{a + b\alpha + c\alpha^2 : a, b, c \in \mathbb{Q}\}.$$

The symmetries of this field extension over $\mathbb{Q}$ are uniquely defined by the image of α among $\alpha, 1 - \frac{1}{\alpha}$ and $\frac{1}{1-\alpha}$. Hence, the symmetries form the group $\mathbb{Z}/3\mathbb{Z}$.

In general, if $K \subseteq L$ is a field extension, we can define

$$\mathrm{Gal}(L/K) \coloneqq \{\sigma : L \to L \text{ field isomorphism} \mid \forall a \in K : \sigma(a) = a\}.$$

One goal of the course will be to show that if $K \subseteq L$ is sufficiently nice (finite and Galois), we get a canonical bijection

$$\{\text{intermediate fields in } K \subseteq L\} \longleftrightarrow \{\text{subgroups of } \mathrm{Gal}(L/K)\}.$$

For future reference, in this course all rings will have a 1.

1.2. **Recollection from MAT1100.** Let us recall a couple of facts about fields from MAT1100.

Exercise 1.2. Show that any homomorphism of fields is injective.

Exercise 1.3. Recall that if k is a field, then we can form the *prime subfield*, which is defined as the intersection of all subfields of k. Show that the prime subfield is either $\mathbb{Q}$ or $\mathbb{F}_p$ for some prime $p > 0$.

We remember that $k[x]$ is a principal ideal domain and that two ideals (f) and (g) are equal if and only if f and g differ by multiplication by a unit. Furthermore, the prime ideals of $k[x]$ are the zero ideal and the ideals (f) where f is a monic irreducible polynomial. Only the latter are maximal.

We also had the factor theorem, telling us that if $\alpha \in k$ and $f(\alpha) = 0$ for some polynomial $f \in k[x]$, then $(x - \alpha)$ divides $f(x)$. As a corollary, we could deduce that a degree n polynomial f has at most n roots.

Date: Winter 2026.

1.3. **Field extensions.** We can now treat the first important definition. A *field extension* is a homomorphism of fields $i : k \to K$, which will often be denoted as K/k, suppressing the map i which is often clear from context. Note that we avoid working with inclusions as this is not a categorical notion. Furthermore, we will often want to think of k as being embedded in K in more than one way.

We remark that i makes K into a vector space over k and we will denote its dimension by $[K : k]$, which is called the *degree of the field extension.* If $[K : k] < \infty$ we shall say that K/k is a finite field extension.

Example 1.4. Some examples of field extensions are

$$\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C} \qquad k \subseteq k(t) \qquad \mathbb{Q} \subseteq \mathbb{Q}(i).$$

The finite field extensions among these are $\mathbb{R} \subseteq \mathbb{C}$ and $\mathbb{Q} \subseteq \mathbb{Q}(i)$.

Whenever we define a new object in mathematics, we are morally obligated to discuss the morphisms. Given two field extensions $k \to K$ and $k \to L$ we say that a ring homomorphism $\varphi : K \to L$ is a *k-homomorphism* if the following diagram commutes:

$$\begin{tikzcd} K \arrow[r, "\varphi"] & L \\ k \arrow[u] \arrow[ur] & \end{tikzcd}$$

In particular, it follows from this definition that a k-homomorphism is k-linear as a map between vector spaces over k. We shall denote the collection of k-homomorphisms by $\mathrm{Hom}_k(K, L)$ and the k-automorphisms of K by $\mathrm{Aut}_k(K)$.

Suppose that we have a field extension K/k and $\alpha \in K$. Then we define the evalutation homorphism (which is k-linear) by

$$\begin{aligned} \mathrm{ev}_\alpha : k[x] &\longrightarrow K \\ f(x) &\longmapsto f(\alpha). \end{aligned}$$

Notice that the image of this map is $k[\alpha] \subseteq K$ and that the kernel must be a prime principal ideal of $k[x]$ since $k[x]/\ker(\mathrm{ev}_\alpha) \hookrightarrow K$.

- Case 1: If $\ker(\mathrm{ev}_\alpha) = 0$ then $k[\alpha] \cong k[x]$ and hence α isn't the zero of any nonzero polynomial over k. In this case, we say that α is *transcendental over k*. Examples of this are $t \in k(t)$ over k and $e, \pi \in \mathbb{R}$ over $\mathbb{Q}$.
- Case 2: If $\ker(\mathrm{ev}_\alpha) = (m_\alpha)$ for a unique monic irreducible polynomial $m_\alpha(x)$, then we say that m_α is the *minimal polynomial of α over k* and we say that α is algebraic over k. Note that for any $f \in k[x]$ we have that $f(\alpha) = 0$ if and only if m_α divides f and that $k[\alpha] = k(\alpha)$ is a field.

Exercise 1.5. Prove that if $\alpha \in K$ is algebraic over k, then

$$[k(\alpha) : k] = \dim_k(k[x]/m_\alpha(x)) = \deg(m_\alpha(x)).$$

2. Lecture 2

2.1. **Finite, algebraic and finitely generated field extensions.** In this lecture, we will explore the relationship between different finiteness notions for field extensions. We will start with some obvious remarks.

We begin by remarking that if $\alpha \in K$ is algebraic over k, then $k[x]/(m_\alpha(x))$ is a field extension of k. Conversely, given a monic irreducible polynomial $f(x) \in k[x]$, we see that

$k_f := k[x]/(f(x))$ is a field extension of k of degree $\deg(f)$ and f has the zero $\bar{x} \in k_f$. In particular, by choosing an irreducible factor, we deduce the following proposition.

Proposition 2.1 (Kronecker). *For all nonconstant $f \in k[x]$ there exists a finite field extension K/k for which f has a root in K.*

We will also need the following easy proposition.

Proposition 2.2 (Tower law). *If $L/K/k$ are field extensions, then*

$$[L : k] = [L : K][K : k].$$

Proof. Pick bases (α_i) of L over K and (β_j) of K over k. It is not hard to show that $(\alpha_i \beta_j)$ is a basis of L/k. □

This already gives us some interesting consequences. A classical application is to constructability questions using a compass and an unmarked ruler.

Exercise 2.3. In this exercise, you will prove that given a cube in $\mathbb{R}^3$ cannot be doubled using a compass and ruler Consider the corners of the unit cube in $\mathbb{R}^3$ and note that the coordinates of each of these points are rational and suppose for the sake of contradiction that this cube can be doubled.

(a) At each step, we will consider the field generated by the coefficients of all the thus far constructed points. Show that when adjoining a new point, we get a field extension of degree at most 2.
(b) Show that the degree of the field generated by the coefficients of the thus constructed points over $\mathbb{Q}$ is always a power of 2.
(c) Argue that $x^3 - 2$ is the minimal polynomial of $\sqrt[3]{2}$ using Eisenstein's criteria and Gauss' lemma.
(d) Deduce, using the tower law, that $\sqrt[3]{2}$ is not in the field generated by the coefficients of all thus constructed points and conclude that the cube cannot be doubled.

Let us now introduce a couple of more definitions for field extensions K/k.

- If $K = k(\alpha_1, \ldots, \alpha_n)$ for some $\alpha_i \in K$ we say that K/k is *finitely genereated.*
- If $K = k(\alpha)$ for some $\alpha \in K$ we say that K/k is *simple* or *primitive.*
- If every element of K is algebraic over k we say that K/k is *algebraic.*

The proof of the following useful universal property follows immediately from the universal property of quotients.

Proposition 2.4. *Given a field extension K/k, a polynomial $f(x) \in k[x]$ and any $\alpha \in K$ such that $f(\alpha) = 0$ there exists a unique k-homomorphism*

$$\begin{aligned} \varphi : k_f &\longrightarrow K \\ \bar{x} &\longmapsto \alpha. \end{aligned}$$

We shall see later that the iterative use of this proposition can be used to construct elements of $\mathrm{Gal}(K/k)$.

Having introduced two different but related notions, that of K/k being finite or finitely generated, we ought to compare them. For instance $k(t)$ is finitely generated over k, yet not finite. In general, we have the following.

Proposition 2.5. *The field extension K/k is finite if and only if it is finitely generated and algebraic.*

Proof. Suppose first that K/k is finite. Obviously, K/k is finitely generated. Furthermore, for any $\alpha \in K$, $1, \alpha, \dots, \alpha^d$ must be linearly dependentent, where $d = [K : k]$, thus α is algebraic. This proves the forward direction.

Assume now that K/k is finitely generated and algebraic. Thus, there exist $\alpha_1, \dots, \alpha_n \in K$ such that $K = k(\alpha_1, \dots, \alpha_n)$. We will prove inductively that K is finite. Recall that we showed in the last lecture that primitive algebraic extensions are finite. This means that each extension $k(\alpha_1, \dots, \alpha_k)(\alpha_{k+1})/k(\alpha_1, \dots, \alpha_k)$ is finite (since algebraic over k implies algebraic over $k(\alpha_1, \dots, \alpha_k)$) and by the tower law, K/k is finite, as required. □

Exercise 2.6. By examining the proof of Proposition 2.5, show that if $\alpha_1, \dots, \alpha_n \in K$ are algebraic over k, then $k(\alpha_1, \dots, \alpha_n)/k$ is an algebraic extension.

Exercise 2.7. If $\alpha_1, \dots, \alpha_n \in K$ are algebraic over k, then $k[\alpha_1, \dots, \alpha_n] = k(\alpha_1, \dots, \alpha_n)$.

We will finish this lecture with a last proposition.

Proposition 2.8. *If $L/K/k$ is a tower of field extensions, then*

(1) *L/k is finite if and only if L/k and K/k are finite.*
(2) *L/k is algebraic if and only if L/K and K/k are algebraic.*

Proof. Let $L/K/k$ be a tower of field extensions.

(1) This follows from the tower law.
(2) The forward direction is trivial. Assume that L/K and K/k are algebraic and let $\alpha \in L$ be arbitrary. Since L/K is algebraic, there exist $c_{n-1}, \dots, c_0$ not all equal to zero such that
$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_0 = 0.$$
The trick now, is to consider the field $K' = K(c_{n-1}, \dots, c_0)$. Since K'/k is finitely generated and algebraic, it is a finite extension by Proposition 2.5. Furthermore, $K'(\alpha)/K'$ is finite. By the tower law, $K'(\alpha)/k$ is finite and thus α is algebraic over k. It follows that L/k is algebraic, as required.

□

The slogan in the above proof is, as is often the case in algebra, to throw away all but finitely much data.

3. Lecture 3

3.1. **Lemmas.** Let us first recall some easy facts.

Exercise 3.1. Let K/k and L/k be field extensions and $\varphi : K \to L$ be a k-homomorphism. Suppose that $f(x) \in k[x]$ and $\alpha \in K$ is a root of f. Then $\varphi(\alpha) \in L$ is also a root of f.

Lemma 3.2. *If K/k is algebraic and $\varphi : K \to K$ is a k-homomorphism, then φ is an automorphism.*

Proof. The slogan of this proof is to zoom in on only finitely much data. As a map of fields we already know that φ is injective. Let $\alpha \in K$ be arbitrary. Then, by the previous exercise,
$$X := \{\beta \in K : m_\alpha(\beta) = 0\}$$
is preserved by φ. Since X is finite and φ is injective, we have that φ is a permutation of X. It follows that α is in the image of φ and since α was arbitrary, φ is an automorphism. □

The above result obviously fails if K/k is not algebraic. Consider for example the non-surjective $\mathbb{Q}$-homomorphism

$$\begin{aligned} \varphi : \mathbb{Q}(t) &\longrightarrow \mathbb{Q}(t) \\ t &\longmapsto t^2. \end{aligned}$$

Exercise 3.3. If K/k is a field extension, then

$$L := \{\alpha \in K : \alpha \text{ is algebraic over } k\}.$$

We remark that this gives in particular that

$$\bar{\mathbb{Q}} := \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$$

is a field. In particular, it is an example of an *algebraic closure*.

3.2. **Algebraic closure.** The goal in this section is to show that, for any field k, there exists a maximal algebraic extension $\bar{k}/k$ and that this is unique up to k-isomorphism. However, it will not be unique up to unique isomorphism, meaning that there is no universal property for $\bar{k}$. This nonuniqueness is measured by the automorphism group $\mathrm{Gal}(\bar{k}/k)$ and is what makes Galois theory interesting.

We see that if $\bar{k}$ exists, then the only irreducible polynomials in $\bar{k}[x]$ are linear.

A field K is called *algebraically closed* if it has any of the following equivalent properties:

(i) Every algebraic extension of K is trivial.
(ii) Every irreducible polynomial of $K[x]$ has degree 1.
(iii) Every non-constant polynomial in $K[x]$ has a root in K.
(iv) Every polynomial in $K[x]$ factors linearly over K.

As examples, we see that $\bar{\mathbb{Q}}$ and $\mathbb{C}$ are algebraically closed, while $\mathbb{R}$ and $\mathbb{F}_p$ are not.

An *algebraic closure of* k is an algebraic field extension $\bar{k}/k$ such that $\bar{k}$ is algebraically closed.

Theorem 3.4. *Any field k has an algebraic closure $\bar{k}/k$.*

Proof. Consider the massive ring

$$A := k[T_{f,i} : f \text{ is monic irreducible and } 1 \leq i \leq \deg(f)].$$

This is an attempt to add in all the missing zeroes. To get a field, we should quotient by a maximal ideal and to get $T_{f,i}$ to be zeroes of f we should quotient by an ideal containing the ideal I which is generated by all coefficients of polynomials

$$f(X) - \prod_i (X - T_{f,i})$$

To find a maximal ideal, all we need to do is verify that I is a proper ideal. Suppose not, then there exist finitely many $a_j \in A$ and c_j coefficients of polynomials $f_j(X) - \prod_i (X - T_{f_j,i})$ such that

$$1 = \sum_j a_j c_j.$$

Using Kronecker's theorem, we can by passing to a finite field extension K/k find zeroes $\alpha_{j,i} \in K$ such that

$$f_j(X) = \prod_i (X - \alpha_{j,i}).$$

If we order the zeroes of f_j consistently, we get a k-homomorphism

$$\begin{aligned}\varphi : A[X] &\longrightarrow K[X] \\ T_{f_j,i} &\longmapsto \alpha_{j,i}.\end{aligned}$$

Applying this to $1 = \sum_j a_j c_j$ we obtain that $1 = 0$ which cannot hold in a field K. Consequently, I is a proper ideal. By Zorn's lemma, there exists a maximal ideal $I \subseteq \mathfrak{m}$.

We can now choose

$$\bar{k} := A/\mathfrak{m}.$$

We claim that this is an algebraic extension. This follows from the fact that the generators of A are algebraic by construction and any extension with algebraic generators is algebraic.

To show that $\bar{k}$ is algebraically closed, note that any polynomial $f(x) \in k[x]$ splits linearly over $\bar{k}$ by construction. For a general $g(x) \in \bar{k}[x]$ we can find a zero $\alpha \in K$ in some finite extension $K/\bar{k}$ and then $g(x) \mid m_{\alpha,k}$. Since $m_{\alpha,k}(x)$ splits over K, then so does $g(x)$. □

Exercise 3.5. Prove that if $k[x_i : i \in I]/k$ is a field extension and if each x_i is algebraic, then $k[x_i : i \in I]/k$ is an algebraic extension. *Hint:* Use the corresponding fact when I is finite.

4. Lecture 4

4.1. **Algebraic closures.** In this lecture, we will prove the uniqueness up to isomorphism of the algebraic closure of a field k.

Proposition 4.1. *Let K/k and L/k be field extensions.*

(i) If K/k is algebraic and L/k is algebraically closed, then there exists a k-morphisms $K \to L$.

(ii) If K/k and L/k are algebraic closures, then there exists a k-isomorphism $K \cong L$.

Proof. Suppose first that K/k is algebraic and that L/K is algebraically closed. Consider the set

$$X := \{(F, \theta) : k \hookrightarrow F \overset{\theta}{\hookrightarrow} K\}.$$

This is clearly nonempty and we can give a partial order by $(F_1, \theta_1) \leq (F_2, \theta_2)$ whenever $F_1 \subseteq F_2$ and $\theta_1 = \theta_2|_{F_1}$. Notice that any chain has an upper bound, since if $(F_i, \theta_i)_{i\in I}$ has an upper bound $(F, \theta) \in X$ with

$$F = \cup_i F_i, \qquad \theta(x) = \theta_i(x) \text{ for } x \in F_i.$$

By Zorn's lemma, there exists a maximal element $(F, \theta) \in X$.

We claim that $F = K$, which would finish the proof. If not, then there exists $\alpha \in K \setminus F$. Since α is algebraic over k it satifies a minimal polynomial m_α and this polynomial has a zero $\beta \in L$ because L is algebraically closed. Hence, by an earlier universal property there is a k-homomorphism $F(\alpha) \to L$, which is a contradiction. This proves (i).

In particular, if K/k and L/k are algebraic closures then there is a k-homomorphism $K \to L$. Since L/k is algebraic, L/K is algebraic and thus this gives us that $K \cong L$ as required, because K is algebraically closed. □

We remark the following:

- If K/k is such that K is algebraically closed, then $L := \{\alpha \in K : \alpha \text{ algebraic over } k\}$ is the algebraic closure of k.
- If $k \subseteq K \subseteq \bar{k}$, then $\bar{K} = \bar{k}$.
- If $k \subseteq K \subseteq \bar{K}$ and K/k is algebraic, then $\bar{k} = \bar{K}$.

4.2. **Splitting fields.** Let K/k be a field extension. If $f(x) \in k[x]$ can be factored linearly in $K[x]$, we say that f *splits* over K. In case f splits over K and $K = k(\alpha_1, \ldots, \alpha_n)$, where $\alpha_1, \ldots, \alpha_n \in K$ are the zeroes of f, we shall call K/k a splitting field of f.

We remark that:

- K/k is a splitting field if and only if it is minimal in the sense that $k \subseteq L \subseteq K$ and f splits over L implies that $L = K$.
- If f splits over L/k, then L contains a unique splitting field of f.
- Splitting fields are finite.
- If f splits over K and $g \mid f$ in $k[x]$, then g also spltis over K.

Proposition 4.2. *Let f be a nonconstant polynomial. Then, f admits a splitting field, which is unique up to isomorphism.*

Proof. Since f splits over $\bar{k}$, we see that f has a splitting field inside of $\bar{k}$, which proves existence. Suppose that K_1/k and K_2/k are splitting fields and consider a k-isomorphism $\phi : \bar{K_1} \to \bar{K_2}$. It is obvious that $\phi(K_1) \subseteq K_2$. Similarly $\phi^{-1}(K_2) \subseteq K_1$, implying $K_2 = \phi(K_1)$. Thus, $\phi|_{K_1}$ is an isomorphism between the splitting fields. $\square$

Example 4.3. A splitting field of $x^3 - 2$ is $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, where $\zeta_3 = e^{2\pi i/3}$. We have the tower

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \zeta_3).$$

Since the first extension has degree 3 and the second extension has degree 2, it follows from the tower law that $[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}]$.

Exercise 4.4. Let f be a polynomial of degree n. Prove that a splitting field has degree dividing $n!$.

An algebraic extension K/k will be called *normal* if any irreducible polynomial $f(x) \in k[x]$ with a root in K splits over K.

Theorem 4.5. *Let K/k be a finite extension. Then, the following are equivalent:*

(i) K/k is normal,
(ii) K is a splitting field of some $f(x) \in k[x]$,
(iii) any k-homomorphism $\phi : K \to \bar{K}$ satisfies that $\phi(K) = K$.

Proof. Suppose that the finite extension K/k is normal and is thus of the form $K = k(\alpha_1, \ldots, \alpha_n)$. We, see that K is a splitting field of $m_{\alpha_1}(x) \cdots m_{\alpha_n}(x)$. This shows that $(i) \implies (ii)$.

Suppose that K is a splitting field of a nonconstant $f(x) \in k[x]$ and consider a k-homomorphism $\phi : K \to \bar{K}$. Then, $\phi(K) = k$, since $K[x]$ is a UFD. This proves $(ii) \implies (iii)$.

Suppose that any k-homomorphism $\phi : K \to \bar{K}$ satisfies that $\phi(K) = K$. Suppose that an irreducible $f(x) \in k[x]$ has a root $\alpha \in K$ and let $\beta \in \bar{K}$ be any other root. Then, we have a map

$$\begin{aligned} k(\alpha) &\longrightarrow \bar{K} \\ \alpha &\longmapsto \beta \end{aligned}$$

by a universal property. By an earlier extension proposition, we deduce that there is a map $\theta : K \to \bar{K}$ such that $\theta(\alpha) = \beta \in K$, which finishes the proof of $(iii) \implies (i)$. $\square$

5. Lecture 5

5.1. **Normal closures.** Let us recall some things from last lecture.

Example 5.1. The field extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not *normal* as $x^3 - 2$ does not split over $\mathbb{Q}(\sqrt[3]{2})$. Another way to see the failure of normality is to observe that we have a $\mathbb{Q}$-homomorphism

$$\begin{aligned} \varphi : \mathbb{Q}(\sqrt[3]{2}) &\longrightarrow \bar{\mathbb{Q}} \\ \sqrt[3]{2} &\longmapsto \sqrt[3]{2} \cdot \zeta_3, \end{aligned}$$

which does not satisfy that $\varphi(\mathbb{Q}(\sqrt[3]{2})) = \mathbb{Q}(\sqrt[3]{2})$.

We observe that normality does not behave as nicely as some other properties under towers of field extensions. However, we do have the following:

Lemma 5.2. *Let $L/K/k$ be field extensions. If L/k is normal, then L/K is normal.*

Proof. Let $\alpha \in L$. Then $m_{\alpha,K} \mid m_{\alpha,k}$ and $m_{\alpha,k}$ splits over L as L/k is normal. Since $L[x]$ is a UFD, this implies that $M_{\alpha,K}$ splits over L. Since $\alpha \in L$ was arbitrary, L/K is normal. □

Example 5.3. Of course, K/k need not be normal in the above lemma. An example would be:

$$\mathbb{Q}(\sqrt[3]{2}, \zeta_3) \supset \mathbb{Q}(\sqrt[3]{2}) \supset \mathbb{Q}.$$

It is not true either that if L/K and K/k are normal, then L/k is normal. An example is given by

$$\mathbb{Q}(\sqrt[4]{2}) \supset \mathbb{Q}(\sqrt{2}) \supset \mathbb{Q}.$$

Given an extension K/k a *normal closure* is an algebraic extension N/K such that N is normal over k and if $N/M/K$ and M is normal over k, then $N = M$. In other words, N is a minimal normal closure of K.

Proposition 5.4. *If K/k is finite, then a normal closure exists and is unique up to K-isomorphism. The normal closure is finite over k.*

Proof. Choose α_i such that $K = k(\alpha_1, \ldots, \alpha_n)$ and consider the polynomial

$$f(x) = \prod_i m_{\alpha_i,k}(x).$$

We claim that a normal closure of K/k is simply a splitting field of f over K. Since we have already proved that splitting fields are finite, exist and are unique up to isomorphism it suffices to prove this claim.

The proof follows from the following two observations.

(1) If $M/K/k$ is such that M/k is normal, then $f(x)$ splits over M by normality and thus M contains a splitting field of f over K.
(2) If M/K is a splitting field of f over K, then M is a splitting field over k (since k(roots of f) $= K$(roots of f)) and thus normal.

□

Example 5.5. The normal closure of $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is unsurprisingly $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$.

5.2. **Seperable extensions.** Consider an algebraic extension K/k. An element $\alpha \in K$ is separable over k if $m_{\alpha,k}$ has distinct roots over a closure. If all elements of K are separable, we say that K/k is a *separable extension*.

Luckily for us, most extensions we will be interested in are separable. The next example is in a precise way essentially the only way that a nonseparable finite extension can arise.

Example 5.6. Consider the extension

$$K = \mathbb{F}_p(t^{1/p}) \supset \mathbb{F}_p(t) = k.$$

Notice that $t^{1/p}$ is not separable since it has minimal polynomial $x^p - t$ over k. To see this, note that $\mathbb{F}_p[t]$ is a UFD with fraction field k and use Eisenstein's criterion and Gauss' lemma. Observe that over $\bar{k}$,

$$x^p - t = (x - t^{1/p})^p$$

since k has characteristic p. Consequently, K/k is not separable.

Recall that if k is a field of characteristic p, then the *Frobenius homomorphism* is a $\mathbb{F}_p$-homomorphism defined by

$$\begin{aligned} \varphi : k &\longrightarrow k \\ a &\longmapsto a^p. \end{aligned}$$

Furthermore, we can define a formal derivative of polynomials in $k[x]$.

Lemma 5.7. *A polynomial $f(x) \in k[x]$ has repeated roots over $\bar{k}$ if and only if f and f' have a common factor.*

Proof. Over $\bar{k}$ we can write $f = c\prod_i (x - \alpha_i)$. Then, by the product rule,

$$f'(\alpha_j) = c\prod_{i \neq j}(\alpha_j - \alpha_i).$$

Hence, we see that f and f' have common factors if and only if f has a repeated root. $\square$

Some easy consequences follow below.

Lemma 5.8. *If $f(x) \in k[x]$ is irreducible, then f has repeated roots if and only if $f' = 0$.*

Lemma 5.9. *If $\mathrm{char}(k) = 0$, then any algebraic extension of k is separable.*

Lemma 5.10. *If $\mathrm{char}(k) = p > 0$ and $f(x) \in k[x]$ is irreducible, then there exists $r \geq 0$ and an irreducible $g(x) \in k[x]$ without repeated roots such that $f(x) = g(x^{p^r})$. The number p^r is the number of times each root of f appears in its factorisation.*

Proof. The proof is by induction on the degree. The base case is trivial. If f has no repeated roots, take $r = 0$ and $g = f$. If f has repeated roots, then $f' = 0$ and thus f has only terms of degree divisible by p. Hence, there exists $f_1 \in k[x]$ such that $f(x) = f_1(x^p)$. It follows that f_1 has strictly smaller degree than f and thus by the induction hypothesis, there exists an irreducible $g(x) \in k[x]$ with no repeated roots such that $f_1(x) = g(x^{p^r})$ for some $r \geq 0$. Hence, $f(x) = g(x^{p^{r+1}})$, and we are done. The last remark of the lemma follows from:

$$g(x) = c\prod(x - \alpha_i) \implies f(x) = c\prod(x^{p^r} - \alpha_i) = c\prod\left(x - \alpha_i^{1/(p^r)}\right)^{p^r}.$$

$\square$

6. Lecture 6

6.1. Perfect fields. Recall that we saw in the last lecture that any algebraic extension of a field of characteristic 0 is separable.

Definition 6.1. We say that a field k is *perfect*, if any algebraic extension K/k is separable.

In particular, it follows that any field of characteristic 0 and any algebraically closed field is perfect. What about general fields of positive characteristic?

Proposition 6.2. *If* $\mathrm{char}(k) = p > 0$*, then* k *is perfect if and only if the Frobenius map* $\varphi : a \mapsto a^p$ *is an isomorphism.*

Proof. Assume first that φ is surjective an let $f(x) \in k[x]$ be an arbitrary irreducible polynomial. By Lemma 5.10 there exists an irreducible polynomial $g(x) \in k[x]$ with distinct zeroes, such that $f(x) = g\left(x^{p^r}\right)$ for some $r \geq 0$. Suppose that

$$g(x) = \sum_i b_i x^i.$$

Since the Frobenius map is an isomorphism, there exists elements $b_i^{1/p^r} \in k$ and thus

$$f(x) = \sum_i b_i x^{p^r i} = \left(\sum_i b_i^{1/p^r} x^i\right)^{p^r}.$$

Since f is irreducible, $r = 0$, which implies that $f = g$ has distinct zeroes. This establishes one of the directions.

Assume now that k is perfect. Let $\alpha \in k$ be arbitrary. Consider the polynomial $x^p - \alpha \in k[x]$. This polynomial has a zero $\beta \in \bar{k}$ and thus by Frobenius:

$$x^p - \alpha = (x - \beta)^p.$$

The minimal polynomial of β must divide the above and since $\bar{k}/k$ is separable, we conclude that the minimal polynomial of β is $x - \beta$. This shows that $\beta \in k$. Hence φ is surjective. □

Example 6.3. The above theorem proves that any finite field k is perfect, since the Frobenius map $\varphi : k \to k$ is an injective map from a set to itself and thus a bijection.

If $k = \mathbb{F}_p(t)$, then $t \notin \varphi(\mathbb{F}_p(t)) = \mathbb{F}_p(t^p)$, which explains why we found a nonseparable extension of k in the last lecture.

6.2. Separable degree. In this subsection, we will show that separable extensions obey a tower law, similar to that for degrees of finite extensions. First, we begin with an easy exercise.

Exercise 6.4. Show that for any tower of algebraic extensions $L/K/k$, if L/k is separable, then L/K and K/k are also separable.

Our goal is to prove the converse of the above exercise.

Definition 6.5. Let K/k be an algebraic extension. Then, we define the *separable degree* of K/k as

$$[K : k]_s := \left|\mathrm{Hom}_k(K, \bar{k})\right|.$$

Let us start by analysing the situation for simple extensions.

Proposition 6.6. *If $\alpha \in K$ is algebraic over k, then*

$$[k(\alpha) : k] = \begin{cases} [k(\alpha) : k]_s & \text{if } \operatorname{char}(k) = 0, \\ p^r [k(\alpha) : k]_s & \text{if } \operatorname{char}(k) = p \text{ and } f(x) = g\left(x^{p^r}\right) \text{ like in lemma.} \end{cases}$$

In particular $[k(\alpha) : k]_s \mid [k(\alpha) : k]$ with equality if and only if α is separable over k.

Proof. Notice that $[k(\alpha) : k]_s$ is exactly the number of distinct zeroes of $m_{\alpha,k}$ in $\bar{k}$. □

We can now state and prove the tower law for separable extensions.

Proposition 6.7. *If $L/K/k$ are finite extensions, then $[L : k]_s = [L : K]_s \cdot [K : k]_s$.*

Proof. Let

$$\operatorname{Hom}_K(L, \bar{K}) = \{\sigma_i\},$$
$$\operatorname{Hom}_k(K, \bar{K}) = \{\tau_j\}.$$

By Proposition 4.1, we can extend each τ_j to an isomorphism $\widetilde{\tau_j} : \bar{K} \to \bar{K}$ which restricts to τ_j on K. Consider now the map:

$$F : \operatorname{Hom}_K(L, \bar{K}) \times \operatorname{Hom}_k(K, \bar{K}) \longrightarrow \operatorname{Hom}_k(L, \bar{K})$$
$$(\sigma_i, \tau_j) \longmapsto \widetilde{\tau_j} \circ \sigma_i.$$

We will first demonstrate that F is injective. By restricting to K we see that by restricting to K,

$$\widetilde{\tau_{j_1}} \circ \sigma_{i_1} = \widetilde{\tau_{j_2}} \circ \sigma_{i_2} \implies \widetilde{\tau_{j_1}}|_K = \widetilde{\tau_{j_2}} \implies \tau_{j_1} \implies \tau_{j_2}.$$

By injectivity of the τ_j, this implies that $\sigma_{i_1} = \sigma_{i_2}$. Hence, the map F is injective.

Suppose now that $\psi : L \to \bar{K}$ is an arbitrary k-homomorphism. Notice that $\psi|_K : K \to \bar{K}$ is a k-homomorphism and consequently there exists a τ_j such that $\psi|_K = \tau_j$. This shows in particular that

$$(\widetilde{\tau_j})^{-1} \circ \psi$$

is a K-homomorphism. Hence, there exists a σ_i such that $\psi = \widetilde{\tau_j} \circ \sigma_i$, proving that F is surjective. In conclusion, the tower law for separable degrees follows. □

This gives us a generalisation of Proposition 6.6.

Proposition 6.8. *Let K/k be a finite extension. Then:*

(i)

$$[K : k] = \begin{cases} [K : k]_s & \text{if } \operatorname{char}(k) = 0, \\ p^r [K : k]_s & \text{if } \operatorname{char}(k = p) \text{ for some } r \geq 0. \end{cases}$$

(ii) K/k is separable if and only if $[K : k] = [K : k]_s$.

Proof. Part (i) follows from Proposition 6.6 and the tower law of degrees by induction.

Suppose first that K/k is separable. Then, by Exercise 6.4, we have $K = k(\alpha_1, \dots, \alpha_n)$, with each

$$k(\alpha_1, \dots, \alpha_{k+1})/k(\alpha_1, \dots, \alpha_k)$$

separable. Proposition 6.6 and the tower law implies that $[K : k] = [K : k]_s$.

For the more interesting direction, suppose that $[K : k]_s = [K : k]$. Let $\beta \in K$ be arbitrary. Then, from equality of degrees, we have that $K/k(\beta)/k$ gives

$$[k(\beta) : k]_s = [k(\beta) : k].$$

Then, Proposition 6.6 shows that $\beta \in K$ is separable over k. Hence, we are done. □

An easy corollary follows from this:

Corollary 6.9. *For finite field extensions $L/K/k$ we have that L/K and K/k are both separable if and only if L/k is separable.*

7. Lecture 7

7.1. Galois theory.

Proposition 7.1 (Primitive element theorem)**.** *If K/k is a finite and separable extension, then there exists $\alpha \in K$ such that $K = k(\alpha)$.*

Proof. There are two cases.

- **Case 1:** Suppose that k is finite field. Then K is also finite and hence it has a multiplicative group $K^\times$ that is cyclic, generated by some $\alpha \in K$. It is clear that $K = k(\alpha)$.
- **Case 2:** Suppose that k is infinite. By induction, we may assume that $K = k(\alpha, \beta)$, where α and β are algebraic and separable over k. Consider an element $\gamma = \alpha + \lambda\beta$. Let f be the minimal polynomial of α and g be the minimal polynomial of β and consider a field extension L/K where these split. Let
$$h(x) := f(\gamma - \lambda x)$$
and notice that $h(\beta) = 0$. Notice that the minimal polynomial of β in $F(\gamma)$ divides both h and g. We want to show that for most choices of λ, the greatest common divisor of h and g is at most linear. Suppose not. Then, they share another root $\beta' \neq \beta$ (since g is separable). Thus, there exists a root α' of f for which $\gamma - \lambda\beta' = \alpha'$. We deduce that
$$\lambda = \frac{\alpha' - \alpha}{\beta - \beta'}.$$
Hence there are only finitely many $\lambda \in k$ such that $K \neq k(\gamma)$ and since k is infinite, we are done.

□

We shall say that an algebraic extension K/k is *Galois* if it is normal and separable. We remark that if $L/K/k$ is a pair of field extensions with L/k Galois, then we can only deduce that L/K is Galois.

Proposition 7.2. *If K/k is finite, then $|\mathrm{Aut}_k(K)| \leq [K : k]$ with equality if and only if K/k is Galois.*

Proof. Notice that
$$|\mathrm{Aut}_k(K)| \leq |\mathrm{Hom}_k(K, \bar{K})| = [K : k]_s \leq [K : k].$$
The first inequality holds if and only if K/k is normal and the second inequality holds if and only if K/k is separable. □

We can now state the *Galois correspondence*. Fix a finite Galois extension K/k. We have maps
$$\{\text{intermediate fields in } K/k\} \rightleftarrows \{\text{subgroups of } \mathrm{Gal}(K/k)\}$$
$$\Theta : F \mapsto \mathrm{Gal}(K/F)$$
$$K^H := \{\alpha \in K : h(\alpha) = \alpha, \forall h \in H\} \leftarrow\!\shortmid H : \Phi$$

Our goal is to show that Θ and Φ are inverse bijections. Let us start with some basic observations.

(i) Both Θ and Φ reverse inclusions.
(ii) $\Theta(\Phi(H)) \supseteq H$ and $\Phi(\Theta(F)) \supseteq F$.
(iii) $\Theta(\Phi(\Theta(F))) = \Theta(F)$ and $\Phi(\Theta(\Phi(H))) = \Phi(H)$ follow from (i) and (ii).
(iv) If F is any intermediate field then K/F is Galois and hence

$$|\Theta(F)| = |\mathrm{Gal}(K/F)| = [K : F].$$

(v) $|\Theta(\Phi(\Theta(F)))| = |\Theta(F)|$ and hence by (iv), $[K : \Phi(\Theta(F))] = [K : F]$.

The tower law, (ii) and (v) imply together that $\Phi(\Theta(F)) = F$. In particular, θ is injective and we obtain the following proposition.

Proposition 7.3. *A finite Galois extension has only finitely many intermediate extensions. By passing to a normal closure, one can extend this result to finite separable extensions.*

We are now left with proving that $\Theta(\Phi(H)) = H$, i.e. $\mathrm{Gal}(K/K^H) = H$.

Proposition 7.4. *Suppose that K is any field and $G \leq \mathrm{Aut}(K)$ is any finite subgroup. Then K/K^G is finite Galois and $\mathrm{Gal}(K/K^G) = G$.*

Proof. Pick $\alpha \in K$ and let

$$f(x) = \prod_{\beta \in G\alpha} (x - \beta) \in K[x].$$

Clearly the coefficients of f are G-invariant and thus $f(x) \in K^G[x]$. It follows that α is algebraic. We claim that f is the minimal polynomial of α over K^G. Clearly $m_\alpha(x) \mid f(x)$. Furthermore, since $x - \alpha$ divides $m_\alpha(x)$ we see by applying g that $x - g(\alpha)$ divides $m_\alpha(x)$ for any $g \in G$. This shows that $m_\alpha = f$. Of course, f is normal and separable in K over K^G. Consequently K/K^G is Galois. Notice that for each $\alpha \in K$:

$$[K^G(\alpha) : K^G] = |G\alpha| \leq |G|.$$

By the primitive element theorem, for every finite subextension $K/F/K^G$, $[F : K^G] \leq |G|$. By taking F to be maximal, it is easy to see by adjoining one element that $F = K$. Hence K/K^G is finite and Galois.

But G embeds into $\mathrm{Gal}(K/K^G)$ and the above shows that $|G| \leq |\mathrm{Gal}(K/K^G)| = [K : K^G] \leq |G|$. Therefore, $G = \mathrm{Gal}(K/K^G)$ as claimed. □

This finishes the proof of the Galois correspondence.

Corollary 7.5. *If K/k is finite and Galois, then for all $\alpha \in K$*

$$m_{\alpha,k}(x) = \prod_{\beta \in \mathrm{Gal}(K/k)\alpha} (x - \beta).$$

Proof. Using that $k = K^{\mathrm{Gal}(K/k)}$. □

Corollary 7.6. *If K/k is finite and $G = \mathrm{Aut}(K/k)$, then K/k is Galois if and only if $K^G = k$.*

Proof. If K/k is Galois, the result follows from the Galois correspondence. If, on the other hand $K^G = k$, then $K/K^G = K/k$ is Galois by a proposition above. □

Theorem 7.7 (The Fundamental Theorem of Galois Theory)**.** *Let K/k be finite and Galois.*

$$\{\textit{intermediate fields in } K/k\} \rightleftarrows \{\textit{subgroups of } \mathrm{Gal}(K/k)\}$$
$$\Theta : F \mapsto \mathrm{Gal}(K/F)$$
$$K^H := \{\alpha \in K : h(\alpha) = \alpha, \forall h \in H\} \leftarrow\!\shortmid H : \Phi$$

(1) The maps Θ and Φ are inverse bijections and inclusion reversing.
(2) If F corresponds to H, then $[K : F] = |H|$.
(3) F/k is Galois if and only if $H \trianglelefteq \mathrm{Gal}(K/k) = G$ and $\mathrm{Gal}(F/k) \cong G/H$.

We will prove part (iii) of the theorem in the next lecture.

Exercise 7.8. If $F_2 \subseteq F_1$ corresponds to $H_1 \leq H_2$, then F_1/F_2 is normal if and only if $H_1 \trianglelefteq H_2$ and in that case $\mathrm{Gal}(F_1/F_2) \cong H_2/H_1$.

8. Lecture 8

8.1. **Finishing the proof of FTGT.** In this lecture we will finish the proof of the fundamental theorem of Galois theory. Firstly, we will state some easy observations. If F_1 and F_2 are fields corresponding to groups H_1 and H_2 under the Galois correspondence, then $F_1 \cap F_2$ corresponds to $\langle H_1, H_2\rangle$ and the compositum F_1F_2 corresponds to $H_1 \cap H_2$.

Let K/k be finite and Galois.

Lemma 8.1. *Suppose that F is any intermediate field of K/k, then F/k is normal/Galois if and only if $\sigma(F) = F$ for all $\sigma \in \mathrm{Gal}(K/k)$.*

Proof. Suppose that F/k is normal and let $\sigma \in \mathrm{Gal}(K/k)$. Then the map

$$\sigma|_F : F \to K$$

must satisfy that $\sigma_F(F) = \sigma(F)$ by one of the equivalent properties of normality.

For the other direction, suppose that $\sigma(F) = F$ for all $\sigma \in \mathrm{Gal}(K/k)$. Consider any map $\varphi : F \to \bar{K}$. Since $\bar{K}$ is algebraically closed, we can extend φ to a map $\varphi : K \to \bar{K}$. As K/k is normal, $\varphi(K) = K$ and hence we obtain a map $\varphi : K \to K$ and thus $\varphi(F) = F$ by assumption. □

Exercise 8.2. If F corresponds to H under the Galois correspondence, then for any $\sigma \in \mathrm{Gal}(K/k)$, we have that $\sigma(F)$ corresponds to $\sigma H \sigma^{-1}$.

We can now tackle part (iii) of the Galois correspondence.

Proof. Notice that F/k is normal if and only if for all $\sigma \in \mathrm{Gal}(K/k)$, $\sigma(F) = F$ by the previous lemma. By Exercise 8.2 this condition is equivalent to $\sigma H \sigma^{-1} = H$. Hence, F/k is normal if and only if $H \trianglelefteq \mathrm{Gal}(K/k)$, as required.

In this case, we have a restriction homomorphism

$$\theta : \mathrm{Gal}(K/k) \longrightarrow \mathrm{Gal}(F/k)$$

with kernel H. Using the tower law we can compute the order of all involved groups to show that θ is surjective. Hence, the first isomorphism theorem implies

$$\mathrm{Gal}(F/k) \cong \frac{\mathrm{Gal}(K/k)}{\mathrm{Gal}(K/F)}.$$

The main point here is of course that the map θ is well-defined by Lemma 8.1. □

Example 8.3. Let us consider an example of what our theory can do for us so far. Let $K = \mathbb{Q}(\sqrt[4]{2}, i)$ be the splitting field of $x^4 - 2$ over $\mathbb{Q}$ and note that $K/\mathbb{Q}$ is finite Galois. By considering the tower

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[4]{2}) \subseteq K$$

and using the tower law, we deduce that $[K : \mathbb{Q}] = 8$. To describe the Galois group $G = \mathrm{Gal}(K/\mathbb{Q})$, we need to specify the image of i and $\sqrt[4]{2}$ which gives at most 8 possibilities. Since $|G| = 8$, all of these give valid $\mathbb{Q}$-automorphisms. By taking

$$\begin{aligned} \sigma : i &\longmapsto i \\ \sqrt[4]{2} &\longmapsto \sqrt[4]{2}i \\ \tau : i &\longmapsto -i \\ \sqrt[4]{2} &\longmapsto \sqrt[4]{2} \end{aligned}$$

It is easy to verify that the relations $\sigma^4 = \tau^2 = 1$ and $\tau\sigma\tau^{-1} = \sigma^{-1}$, which is sufficient to deduce that $G \cong D_8$. The subgroup lattice

$$\begin{tikzcd}
 & & D_8 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & & \\
 & \langle \sigma^2, \sigma\tau \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \langle \sigma \rangle \arrow[d, dash] & \langle \sigma^2, \tau \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle \sigma^{-1}\tau \rangle \arrow[drr, dash] & \langle \sigma\tau \rangle \arrow[dr, dash] & \langle \sigma^2 \rangle \arrow[d, dash] & \langle \tau \rangle \arrow[dl, dash] & \langle \sigma^2\tau \rangle \arrow[dll, dash] \\
 & & 1 & &
\end{tikzcd}$$

gives us the lattice of intermediate fields:

$$\begin{tikzcd}
 & & \mathbb{Q}(\sqrt[4]{2}, i) \arrow[dll, dash] \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\sqrt[4]{2}(1-i)) \arrow[dr, dash] & \mathbb{Q}(\sqrt[4]{2}(1+i)) \arrow[d, dash] & \mathbb{Q}(\sqrt{2}, i) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \mathbb{Q}(\sqrt[4]{2}) \arrow[d, dash] & \mathbb{Q}(\sqrt[4]{2}i) \arrow[dl, dash] \\
 & \mathbb{Q}(i\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(i) \arrow[d, dash] & \mathbb{Q}(\sqrt{2}) \arrow[dl, dash] & \\
 & & 1 & &
\end{tikzcd}$$

8.2. **Finite fields.** Notice that if k is a finite field with prime subfield $\mathbb{F}_p$ and $r = [k : \mathbb{F}_p]$, then by linear algebra $|k| = p^r$.

Proposition 8.4. *Suppose that $\mathbb{F}_q$ is a finite field with $q = p^r$. Then, for each $n \geq$ there exists a unique field extension $k/\mathbb{F}_q$ of degree n up to $\mathbb{F}_q$-isomorphism. Then, $k/\mathbb{F}_q$ is Galois and $\mathrm{Gal}(k/\mathbb{F}_q)$ is cyclic of order n, generated by the Frobenius map $\varphi^r : x \mapsto x^q$.*

Proof. For the proof of the uniqueness, note that if $k/\mathbb{F}_q$ has degree n, then $|k^\times| = q^n - 1$. Thus, $\alpha^{q^n} - \alpha = 0$ for all $\alpha \in k$. Thus, by considering the degree, we see that k is the splitting field of the polynomial $x^{q^n} - x \in \mathbb{F}_q[x]$ and therefore unique up to $\mathbb{F}_q$-isomorphism.

For the proof of the existence, let k be the splitting field of $x^{q^n} - x$ over $\mathbb{F}_q$. Consider

$$k' = \{\alpha \in k : \alpha^{q^n} = \alpha\} = \{\alpha \in k : \varphi^{nr}(\alpha) = \alpha\}$$

which is a subfield of k. Hence, $|k'| \leq q^n$. It remains to show that $|k'| = q^n$, i.e. that $x^{q^n} - x$ has distinct roots. This is easy to check by computing derivatives.

Lastly, we claim that $k/\mathbb{F}_q$ is Galois. It is separable since $\mathbb{F}_q$ is perfect and normal as a splitting field.

Notice that $\varphi^r : \alpha \mapsto \alpha^q$ is an element of $\mathrm{Gal}(k/\mathbb{F}_q)$. It remains to check that this element generates the Galois group. But if $i \geq 1$ and $(\varphi^r)^i = \mathrm{id}$, then $\alpha^{q^i} - \alpha$ has at least $|k|$ zeroes and thus $q^n \leq q^i$ and $i \geq n$. This shows that a power of the Frobenius map generates the Galois group. □

We remark that with $r = 1$ we obtain the special case that there exists a unique finite field of each size q up to field isomorphism.

9. Lecture 9

9.1. **Solvability by radicals.** In this section, we will assume that k is a field of characteristic 0.

Intuitively, we want to say that a polynomial is solvable if the roots can be obtained by $+, -, \cdot, \div$ and $\sqrt[n]{\ }$. How should we formalise this?

Definition 9.1. A finite extension K/k is *radical* if there exist intermediate extensions

$$k = k_0 \subseteq k_1 \subseteq \cdots \subseteq k_r$$

such that $K \subseteq k_r$ and for all $1 \leq i \leq r$, $k_i = k_{i-1}(\alpha_i)$ where $\alpha_i^{n_i} \in k_{i-1}$ for some $n_i \geq 1$.

As an example, $\sqrt[3]{\sqrt{2} + \sqrt[7]{-1}}$ lies in a radical extension. Thus, we shall say that f is a *solvable in radicals* if there exists a radical extension K/k such that f splits over K, or equivalently, any splitting field of f is a radical extension.

Our goal is to show that the group f is solvable if and only if $\mathrm{Gal}(K/k)$ is a solvable group, where K is a splitting field of f over k.

Theorem 9.2. *Let K/k be finite and Galois. Then K/k is radical if and only if* $\mathrm{Gal}(K/k)$ *is solvable.*

Let us recall a few things from group theory.

- A group G is solvable if there exists a chain
$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_r = G$$
with each G_i/G_{i-1} abelian. Equivalently, there exists such a chain with G_i/G_{i-1} cyclic.
- Subgroups and quotient groups of solvable groups are solvable.
- If $N \trianglelefteq G$, then G is solvable if and only if N and G/N are solvable.
- Abelian groups are solvable.
- S_4 is solvable.
- A_5 and S_5 are not solvable.

Definition 9.3. A finite extension K/k is called *abelian* if $\mathrm{Gal}(K/k)$ is abelian and *cyclic* if $\mathrm{Gal}(K/k)$ is cyclic.

Lemma 9.4. *Let K be a splitting field of $x^n - 1$ over k. Then K/k is abelian. Moreover, $\mu_n(k) = \{\zeta \in k : \zeta^n = 1\}$ is cyclic and $\mu_n(K)$ is cyclic of order n.*

Proof. Note that $\mu_n(K) \leq K^\times$ of order exactly n, since $x^n - 1$ is separable because we are in characteristic 0. It follows from the structure theorem for finite abelian groups that there exists a $d \leq n$ such that for all $a \in \mu_n(K)$, $a^d = 1$. Since $x^d - 1$ has at most d roots, it follows that $d = n$ and thus that $\mu_n(K)$ is cyclic of order n.

Let ζ_n be a primitive root of $x^n = 1$. Then, we have that for any $\sigma \in \mathrm{Gal}(K/k)$, there exists an integer $\theta(\sigma) \in (\mathbb{Z}/(n))^\times$ such that

$$\sigma(\zeta_n) = \zeta_n^{\theta(\sigma)}.$$

Notice furthermore that $\sigma : \mathrm{Gal}(K/k) \to (\mathbb{Z}/(n))^\times$ is an injective group homomorphism. Hence, $\mathrm{Gal}(K/k)$ is isomorphic to a subgroup of an abelian group and is hence abelian, as required. □

Example 9.5. If $k = \mathbb{Q}$, then Gauss proved that in fact σ gives an isomorphism

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/(n))^\times.$$

If, on the other hand, $k = \mathbb{C}$, then clearly the Galois group is trivial.

Lemma 9.6. *Suppose that $\zeta_n \in k$. Then, if $K = k(\alpha)$ with $\alpha^n \in k$, it follows that K/k is cyclic.*

Proof. Let $f(x) = x^n - \alpha^n$. Then, the splitting field of f is

$$k(\alpha, \zeta_n) = k(\alpha) = K.$$

Thus, K/k is Galois. Notice also that for any $\sigma \in \mathrm{Gal}(K/k)$, there exists $\theta(\sigma) \in \mathbb{Z}/n\mathbb{Z}$ for which

$$\sigma(\alpha) = \alpha \cdot \zeta_n^{\theta(\sigma)}.$$

It is easy to check that $\theta : \mathrm{Gal}(K/k) \to \mathbb{Z}/n\mathbb{Z}$ is an injective group homomorphism. Thus $\mathrm{Gal}(K/k)$ is isomorphic to a subgroup of the cyclic group $\mathbb{Z}/n\mathbb{Z}$. Hence, it is cyclic. □

Let us now give a nonstandard definition.

Definition 9.7. A field extension M/k is called *polyabelian*, if there exist subextensions

$$k = k_0 \subseteq \cdots k_r = M$$

with each k_i/k_{i-1} abelian.

As an example, $\mathbb{Q}(\sqrt[4]{2}) \supseteq \mathbb{Q}$ is polyabelian, but not even Galois.

By the Fundamental Theorem of Galois Theory, if M/k is polyabelian and Galois, then $\mathrm{Gal}(M/k)$ is solvable.

Lemma 9.8. *If K/k has polyabelian subextensions E/k and F/k, then EF/k is polyabelian. Furthermore, if K/k is polyabelian, then a normal closure N/k is polyabelian.*

Proof. By assumption we have chains

$$k = k_0 \subseteq \cdots \subseteq k_r = E$$
$$k = k'_0 \subseteq \cdots \subseteq k'_s = F$$

for which each k_i/k_{i-1} and k'_i/k'_{i-1} is abelian. It is clear that it is sufficient to show that

$$E = Ek'_0 \subseteq Ek'_s = EF$$

has the property that each Ek_i'/Ek_{i-1}' is abelian to prove the first part of the lemma. Since k_i'/k_{i-1}' is Galois it follows directly that Ek_i'/Ek_{i-1}' is Galois. Furthermore, we have an injective restriction homomorphism:

$$\begin{aligned}\mathrm{Gal}(Ek_i'/Ek_{i-1}') &\longrightarrow \mathrm{Gal}(k_i'/k_{i-1}')\\ \sigma &\longmapsto \sigma|_{k_i'}.\end{aligned}$$

Hence $\mathrm{Gal}(Ek_i'/Ek_{i-1}')$ is abelian.

For the second part of the lemma, let $G = \mathrm{Gal}(N/k)$ and consider

$$M := \langle \sigma(K) : \sigma \in G\rangle.$$

Since each $\sigma(K)/k$ is polyabelian, we deduce that M/k is polyabelian, by the first part of this lemma. Furthermore, M/k is clearly normal since for all $\sigma \in G$, $\sigma(M) = M$. Hence, by definition of the normal closure, $M = N$. Thus, we have proved that N/k is polyabelian if K/k is polyabelian. □

We can now give the proof of the forward direction of Theorem 9.2.

Proof. Assume that K/k is finite, Galois and radical. Then, there exist

$$k = k_0 \subseteq \cdots \subseteq k_r$$

with $K \subseteq k_r$ and for each $1 \leq i \leq r$

$$k_i = k_{i-1}(\alpha_i) \quad \text{where} \quad \alpha_i^{n_i} \in k_{i-1}.$$

Working inside of $\bar{k_r}$, we can find ζ_N with $N = \mathrm{lcm}(n_i)$. Notice that we have

$$k \subseteq k_0(\zeta_N) \subseteq k_1(\zeta_N) \subseteq \cdots \subseteq k_r(\zeta_N) \subseteq N$$

where N is the normal closure of $k_r(\zeta_N)/k$. By Lemma 9.6 and Lemma 9.4, we see that $k_r(\zeta_N)/k$ is polyabelian. By Lemma 9.8 N/k is polyabelian. Since N/k is Galois it follows that $\mathrm{Gal}(N/k)$ is solvable and since this group contains $\mathrm{Gal}(K/k)$ as a quotient group, the latter is also solvable. □

10. Lecture 10

In this lecture, we will prove the backward direction of Theorem 9.2.

10.1. **Correspondence between solvability and radical extensions.**

Lemma 10.1 (Dedekind)**.** *Let F be a field and G any group. Then any distinct characters*

$$\chi_1, \ldots, \chi_n : G \to F^\times$$

are linearly independent.

Proof. Suppose not. Then pick a counterexample $\sum_{i=1}^m \lambda_i\chi_i = 0$ with m smallest. Then for all $g, h \in G$:

$$\sum_{i=1}^m \lambda_i\chi_i(g) = 0,$$

$$\sum_{i=1}^m \lambda_i\chi_i(g)\cdot\chi_i(h) = 0,$$

and taking a linear combination gives

$$\sum_{i=1}^{m} \lambda_i \chi_i(g)(\chi_m(h) - \chi_i(h)) = 0$$

which is a shorter counterexample, unless $\chi_m = \chi_i$. This finishes the proof of the linear independence. □

Lemma 10.2 (Hilbert 90)**.** *Let K/k be cyclic of degree n where $\mathrm{Gal}(K/k) = \langle\sigma\rangle$. If $\alpha \in K^\times$ is such that*

$$\prod_{i=0}^{n-1} \sigma^i(\alpha) = 1,$$

then $\alpha = \beta/\sigma(\beta)$ for some $\beta \in K$.

Proof. By Dedekind, 1, σ, $\dots$, σ^{n-1} are linearly independent as characters $K^\times \to K^\times$. Therefore, the linear combination

$$\mathrm{id} + \alpha \cdot \sigma + \alpha \cdot \sigma(\alpha) \cdot \sigma^2 + \cdots + \alpha \cdot \sigma(\alpha) \cdot \sigma^2(\alpha) \cdots \sigma^{n-2}(\alpha) \cdot \sigma^{n-1}$$

is nonzero. Therefore, there exists $\gamma \in K^\times$ for which

$$\beta = \mathrm{id} + \alpha \cdot \sigma(\gamma) + \alpha \cdot \sigma(\alpha) \cdot \sigma^2(\gamma) + \cdots + \alpha \cdot \sigma(\alpha) \cdot \sigma^2(\alpha) \cdots \sigma^{n-2}(\alpha) \cdot \sigma^{n-1}(\gamma) \neq 0.$$

Notice that $\beta/\sigma(\beta) = \alpha$, as required, using that $\prod_i \sigma^i(\alpha) = 1$. □

Lemma 10.3. *If $\zeta_n \in k$ and K/k is cyclic of degree n. Then, there exists $\alpha \in K$ such that $K = k(\alpha)$ and $\alpha^n \in k$.*

Proof. Since $\mathrm{Gal}(K/k) = \langle\sigma\rangle$ is cyclic, we notice that

$$\prod_i \sigma^i(\zeta_n) = \zeta_n^n = 1$$

and thus by Hilbert 90, there exists $\alpha \in K$ for which $\zeta_n = \alpha/\sigma(\alpha)$. This gives us that $m_{\alpha,k}(x)$ has roots $\alpha \cdot \zeta_n^{-i}$, of which there are n distinct ones. Hence, for degree reasons, $K = k(\alpha)$.

Also,

$$\sigma^i(\alpha^n) = (\alpha \cdot \zeta_n^{-1})^n = \alpha^n$$

and by the Galois correspondence, $\alpha^n \in k$. □

We are now ready to prove the backward direction of Theorem 9.2.

Proof. Assume that K/k is finite Galois and that $\mathrm{Gal}(K/k)$ is solvable. Let $d \coloneqq [K : k]$. By the FTGT, there exist intermediate fields

$$k = k_0 \subset k_1 \subset \cdots \subset k_r = K$$

for which k_i/k_{i-1} is cyclic of degree d_i.

Inside $\bar{K}$,

$$k \subset k_0(\zeta_d) \subset \cdots \subset k_r(\zeta_d).$$

As in homework, $k_i(\zeta_d)/k_{i-1}(\zeta_d)$ is cyclic of degree dividing d_i. By the earlier lemma, there exists α_i such that $\alpha_i^{d_i} \in k_{i-1}(\zeta_d)$ and $k_i(\zeta_d) = k_{i-1}(\zeta_d)(\alpha)$. Also $k_0(\zeta_d)/k_0$ is radical and we deduce that K/k is radical. □

10.2. **Application to solvability in radicals.** Suppose $f(x)$ is irreducible of degree n over a characteristic 0 field k. Let N/k be the splitting field, which is Galois. Then $G = \mathrm{Gal}(N/k)$ acts transitively and faithfully on the set X of roots of f. Thus, G is up to isomorphism a subgroup of S_n. Furthermore, the inclusions

$$k \subseteq k(\alpha) \subseteq N$$

give that $n \mid [N : k]$ by the tower law.

Example 10.4. We claim that the polynomial $f(x) = x^5 - 4x + 2$ over $\mathbb{Q}$ is not solvable by radicals.

Proof. We will show that the galois group G is S_5 in this case, which is not solvable.

By considering the derivative $f'(x) = 5x^4 - 4$ has only two real roots, which shows that f has at most three roots. The fact that

$$\begin{aligned} \lim_{x\to\pm\infty} f(x) &= \pm\infty, \\ f(0) &> 0, \\ f(1) &< 0, \end{aligned}$$

implies that f has exactly three real roots. Hence, complex conjugation $\tau : \mathbb{C} \to \mathbb{C}$ restricts to an element of G corresponding to a 2-cycle.

Furthermore, since 5 divides $|G|$, there exists an element of order 5 by Cauchy's theorem. Since G is a subgroup of S_5, it contains a 5-cycle.

Since a 2-cycle together with a 5-cycle generate S_5, we obtain $G = S_5$. $\square$

- The transitive subgroups of S_5 are C_5, D_{10}, F_{20}, A_5 and S_5 and all are possible as Galois groups.
- For a cyclotomic extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ we have an injection $G = \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/(n))^\times$ and Gauss proved that this is an isomorphism. Equivalently,

$$\Phi_n(x) := \prod_{i\in(\mathbb{Z}/(n))^\times} (x - \zeta_n^i)$$

  is irreducible.

Theorem 10.5 (Kronecker-Weber)**.** *If $K/\mathbb{Q}$ is any finite abelian extension, then it is contained in some $\mathbb{Q}(\zeta_n)$.*

11. Lecture 11

11.1. **Commutative algebra.** In this lecture, we start the commutative algebra part of this course. We will assume throughout that R is a possibly noncommutative ring and that M is a left R-module.

Definition 11.1. An R-module M is *noetherian* if it satisfies the ascending chain condition for submodules. In other words, for every chain of submodules $M_1 \subseteq M_2 \subseteq \cdots$ we have that $M_n = M_{n+1}$ for all n large enough. A ring R is *noetherian* if it is noetherian as an R-module.

Please note that the chains in Zorn's lemma are much more general than the chains in the ascending chain condition.

Lemma 11.2. *The following are equivalent for any R-module M.*

(i) M is noetherian.
(ii) Every submodule of M is finitely generated.

Proof. Suppose (i) and take $N \subseteq M$. If N was not finitely generated we could pick recursively $x_1 \in N \setminus \{0\}$, $x_2 \in N \setminus (x_1), \cdots$ to form an ascending chain

$$0 \subset (x_1) \subset (x_1, x_2) \subset \cdots$$

that never stabilises, which would be a contradiction. Hence (ii) follows from (i).

Assume that (ii) holds. If we have any ascending chain

$$M_1 \subseteq M_2 \subseteq \cdots$$

then the union $N = \bigcup_i M_i$ is again a submodule of M. By (ii), there exist $\alpha_1, \dots, \alpha_n \in M$ such that $N = (\alpha_1, \dots, \alpha_n)$. However, we can pick k large enough such that every α_i is in M_k which implies that the ascending chain stabilises after M_k. Hence (i) follows from (ii). $\square$

Example 11.3. If k is a field, then a k-module is noetherian if and only if it is finite dimensional.

Corollary 11.4. *A ring R is noetherian if and only if all its ideals are finitely generated.*

Example 11.5. Any PID is noetherian. However, $k[x_1, \dots, x_n]$ is not as demonstrated by the ascending chain:

$$(x_1) \subset (x_1, x_2) \subseteq \cdots$$

Lemma 11.6. *If $N \subseteq M$, then M is noetherian if and only if N and M/N are both noetherian.*

Proof. Let $\pi : M \to M/N$ be the projection map. For the forward direction, assume that M is noetherian. We see that any ascending chain of submodules in N is in particular an ascending chain of submodules in M and therefore stabilises. This shows that N is noetherian. Furthermore, for any sequence of submodules of M/N

$$\bar{M_1} \subseteq \bar{M_2} \subseteq \cdots$$

it follows from the correspondence theorem that we have a sequence of corresponding submodules

$$N \subseteq M_1 \subseteq M_2 \subseteq \cdots$$

We know that for k large $M_k = M_{k+1}$ which by the correspondence theorem gives $\bar{M_k} = \bar{M}_{k+1}$.

For the other direction, assume that both N and M/N are noetherian. Consider any ascending chain

$$M_1 \subseteq M_2 \subseteq \cdots$$

of submodules in M. Then, since N and M/N are noetherian it follows that $M_n \cap N = M_{n+1} \cap N$ and $M_n + N = M_{n+1} + N$. We claim that this implies that $M_n = M_{n+1}$.

To prove this, let $x \in M_{n+1}$ be arbitrary. Then there exists $y \in M_n$ and $\nu \in N$ such that $x = y + \nu$. Since $\nu \in M_{n+1} \cap N = M_n \cap N$ it follows that $x - y \in M_n$ which implies that $x \in M_n$, as required. $\square$

Corollary 11.7. *If $M_1, \dots, M_n$ are noetherian R-modules, then $M_1 \oplus \cdots \oplus M_n$ is also noetherian.*

Proof. If M_1 and M_2 are noetherian then $(M_1 \oplus M_2)/M_1 \cong M_2$ and M_1 are noetherian and hence by the previous lemma $M_1 \oplus M_2$ is too. The general claim follows from induction. □

Corollary 11.8. *If R is noetherian and M is an R-module, then M is noetherian if and only if M is finitely generated over R.*

Proof. The forward direction is trivial. For the other direction, assume that M is finitely generated over R. Hence, there exists a surjection $\pi : R^{\oplus n} \to M$ and

$$M \cong R^{\oplus n}/\ker(\pi).$$

As a quotient of a noetherian module we can conclude that M is noetherian. □

11.2. **Hilbert's basis theorem.** From now on, we will assume that R is commutative. The main result of this section is:

Theorem 11.9 (Hilbert's Basis Theorem)**.** *If R is noetherian then $R[x]$ is noetherian.*

Lemma 11.10. *If R is noetherian and I is an ideal of R, then R/I is noetherian as a ring.*

Proof. This is by the correspondence theorem between ideals of R/I and ideals of R containing I. □

Corollary 11.11. *From Hilbert's Basis Theorem and the above lemma one concludes that any finitely generated R-algebra is noetherian as a ring if R is noetherian.*

The above corollary is one of the most useful consequences of Hilbert's basis theorem. We will now proceed with a proof.

Proof. Let R be noetherian and assume for the sake of contradiction that the ideal J of $R[x]$ is not finitely generated. Then, we can recursively pick f_0 to be an element of minimal degree in $J \setminus \{0\}$, f_1 an element of minimal degree in $J \setminus (x_0)$ and so on. Let a_n be the leading coefficient of f_n and let $d_n = \deg(f_n)$.

As R is noetherian, the ascending chain

$$(a_0) \subseteq (a_0, a_1) \subseteq \cdots$$

must stabilise after say $(a_0, \dots, a_{N-1})$. In particular

$$a_N = \sum_i \lambda_{i=0}^{N-1} \lambda_i a_i$$

for some $\lambda_i \in R$. This implies that

$$f_N - \sum_i \lambda_i \lambda_i x^{d-d_i} \cdot f_i(x) \in J \setminus (f_0, \dots, f_{N-1})$$

contradicts the minimality of f_N, which is a contradiction. Hence, J is finitely generated and we are done. □

12. Lecture 12

In this lecture, all rings will be assumed to be commutative. We now give definitions similar to those for field extensions earlier in the course.

Definition 12.1. A ring extension $R \subseteq S$ is called

(i) *finite*, if S is a finitely generated R-module,
(ii) *finite type*, if $S = R[s_1, \dots, s_n]$ for some $s_i \in S$,

(iii) *integral*, if any element of S is a zero of a monic polynomial in $R[x]$.

Example 12.2. Consider the following examples.

- $R \subseteq R[x]$ is finite type, but not integral.
- $\mathbb{Z} \subseteq \mathbb{Q}$ is neither finite type nor integral.
- $k \subseteq k(x)$ is neither finite type nor integral.
- $\mathbb{Z} \subseteq \mathbb{Z}[i]$ is both finite and integral.

Lemma 12.3. *The following are equivalent for a ring extension $R \subseteq S$ of commutative rings and an element $s \in S$.*

(i) s is integral over R,
(ii) $R[s] \subseteq S$ is a finitely generated R-module,
(iii) there exists a subring $R[s] \subseteq R' \subseteq S$ such that R' is a finitely generated R-module.

Proof. If s satisfies a monic polynomial of degree n, then $R[s]$ is spanned as an R-module by $1, s, \ldots, s^{n-1}$ which proves that (i) implies (ii).

The fact that (ii) implies (iii) is trivial.

Assume now that (iii) holds. Hence, there exist $\alpha_1, \ldots, \alpha_n \in R'$ such that $R' = \sum_i R\alpha_i$. In particular, we have

$$s\alpha_i = \sum_j r_{ij}\alpha_j$$

for some matrix $A = (r_{ij})$, from which it follows that

$$(s \cdot \mathrm{id} - A) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = 0.$$

By multiplying by the adjugate, we obtain that

$$\det(s \cdot \mathrm{id} - A) \cdot (\alpha_1, \ldots, \alpha_n) = 0.$$

Since $1 \in R'$, there exist $r_i \in R'$ such that $1 = \sum_i r_i\alpha_i$ and thus $\det(s \cdot \mathrm{id} - A) \cdot 1 = 0$. This gives us a monic polynomial with coefficients in R that is satisfied by s. Hence (i) follows. $\square$

Corollary 12.4. *A ring extension $R \subseteq S$ is finite if and only if it is finite type and integral.*

Proof. The forward direction follows from (iii) implies (i) in Lemma 12.3.

For the background direction, assume that $R \subseteq S$ is finite type. By an argument as in the tower law, we can reduce the proof to an extension of the form $R \subseteq R[s]$. However, it is clear from say Lemma 12.3 that finite type and integral implies finite for such extensions. $\square$

We remark that this shows that if $s_1, \ldots, s_n \in R$ are integral then $R[s_1, \ldots, s_n]$ is also integral.

Corollary 12.5. *Given $R \subseteq S$ we have that the integral closure of R in S $\tilde{R} \coloneqq \{s \in S :$ s is integral over $R\}$ is a subring of S containing R.*

Example 12.6. For any field extension $K/\mathbb{Q}$ we denote the integral closure of $\mathbb{Z}$ by $\mathcal{O}_K \coloneqq \tilde{\mathbb{Z}}$.

Lemma 12.7. *If R is a UFD then $\tilde{R} = R$ in $K \coloneqq \mathrm{Frac}(R)$.*

Proof. Suppose for the sake of contradiction that there exists $\alpha \in \tilde{R} \setminus R$. Then $\alpha = \frac{a}{b}$ with $a, b \in R \setminus \{0\}$ and using the fact that R is a UFD there exists a prime $p \in R$ such that p divides b but not a. Since α is integral it satisfies a monic polynomial:

$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_0 = 0.$$

Clearing denominators we get that

$$a^n + c_{n-1}a^{n-1}b + \cdots + b^n c_0 = 0.$$

We see that p divides all terms except a^n which is a contradiction. □

Example 12.8. Recall that $\mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$ so this shows in particular that $\mathbb{Z}[\sqrt{5}]$ is not a UFD.

Lemma 12.9. *If R is a domain and $\mathrm{Frac}(R) = K \subseteq L$ is a field extension, then for all $\alpha \in L$ that are algebraic over K there exists $r \in R \setminus \{0\}$ such that $r\alpha \in L$ is integral over R.*

Proof. By assumption we have an identity of the form

$$\alpha^n + \lambda_{n-1}\alpha^{n-1} + \cdots + \lambda_0 = 0$$

for some $\lambda_i \in K$. We may assume that each λ_i has the same denominator $r \neq 0$ and multiplying through we obtain that

$$(r\alpha)^n + r\lambda_{n-1}(r\alpha)^{n-1} + \cdots + r^n\lambda_0 = 0$$

and each coefficient above is in R. Thus $r\alpha$ is integral over R as required. □

Theorem 12.10 (Zariski's lemma)**.** *If K/k is a finite type field extension, then it is finite.*

Proof. Suppose that there exist $\alpha_1, \ldots, \alpha_n \in K$ such that $K = k[\alpha_1, \ldots, \alpha_n]$. Th e proof proceeds by induction. If $n = 0$ there is nothing to prove. If $n = 1$ and we have $K = k[\alpha_1]$, then if α_1 is transcendental we have $K \cong k[x]$ which contradicts the fact that K is a field. Thus α_1 is algebraic over k and K is thus finite.

For the induction step, assume that the claim has been proved for $n - 1$. Consider

$$k \subseteq k(\alpha_1) \subseteq K(\alpha_1)[\alpha_2, \ldots, \alpha_n].$$

The second subextension is finite by the induction hypothesis and thus if α_1 is algebraic over k, we are done. Hence, assume for the sake of contradiction that α_1 is transcendental over k and thus $R := k[\alpha_1] \cong k[x]$ which is a UFD. We know that each α_i is algebraic over $k(x) = \mathrm{Frac}(R)$ and thus by the previous lemma there exists $r \neq 0$ such that $r\alpha_i$ is integral over R for each i. Hence, for any $\gamma \in \mathrm{Frac}(R)$ there exists $N \in \mathbb{N}$ such that $r^N\gamma$ is integral over R. However, since R is a UFD this would imply that $r^N\gamma \in R$, since it belongs to the fraction field. This contradicts the fact that there are infinitely many non-associate primes in $k[x]$. Thus we are done. □

13. Lecture 13

13.1. **Weak Nullstellensatz.** We will start by giving an alternative proof of Zariski's lemma that works whenever k is uncountable.

Proof. Suppose that $K = k[\alpha_1, \dots, \alpha_n]$. Then, K is spanned by countably many monomials in α_i. Thus, $\dim_k(K)$ is countable.

Suppose for the sake of contradiction that K/k is not finite. Then, there exists $t \in K$ that is transcendental over k. Notice that

$$\left\{ \frac{1}{t-a} : a \in k \right\}$$

is an uncountable collection of linearly independent vectors in K. This contradicts the fact that $\dim_k(K)$ is countable.

Hence K/K is finite, which was what we wanted to prove. □

As a corollary of Zariski's lemma, we obtain the weak nullstellensatz.

Theorem 13.1 (Weak Nullstellensatz)**.** *Let k be an algebraically closed field. Then, the maximal ideals of $k[x_1, \dots, x_n]$ are exactly the ideals of the form*

$$(x_1 - a_1, \dots, x_n - a_n)$$

for $(a_1, \dots, a_n) \in k^n$.

Proof. Suppose that $a = (a_1, \dots, a_n) \in k^n$ and consider the ideal $(x - a_1, \dots, x_n - a_n)$. Consider the surjective evaluation map

$$\mathrm{ev}_a : k[x_1, \dots, x_n] \longrightarrow k.$$

Then, it is clear that $\ker(\mathrm{ev}_a) = (x_1 - a_1, \dots, x_n - a_n)$ from which it follows that

$$k[x_1, \dots, x_n]/(x_1 - a_1, \dots, x_n - a_n) \cong k$$

which is a field. Consequently, $(x_1 - a_1, \dots, x_n - a_n)$ is maximal. This proves one direction of the theorem.

For the harder direction, assume that $\mathfrak{m}$ is a maximal ideal in $k[x_1, \dots, x_n]$. Consider the map

$$\phi : k \longrightarrow K \coloneqq k[x_1, \dots, x_n]/\mathfrak{m}$$

which clearly exhibits K as a finite type field extension of k. By Zariski's lemma, K/k is finite and since k is algebraically closed, ϕ is an isomorphism. Choose $a_i = \phi^{-1}(x_i)$. Then each $x_i - a_i \in \mathfrak{m}$ and thus

$$(x_1 - a_1, \dots, x_n - a_n) \subseteq \mathfrak{m}.$$

Since $(x_1 - a_1, \dots, x_n - a_n)$ is maximal, it follows that

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n).$$

This finishes the proof. □

13.2. **Nullstellensatz.** Assume in this section that k is algebraically closed. Given an ideal $J \subseteq k[x_1, \dots, x_n]$, we define

$$V(J) \coloneqq \{a \in k^n : f(a) = 0 \text{ for all } f \in J\}$$

and given $X \subseteq k^n$, we define

$$I(X) \coloneqq \{f \in k[x_1, \dots, x_n] : f(a) = 0 \text{ for all } a \in X\}.$$

Our goal is to show that V and I are almost inverses. Any subset of k^n of the form $V(J)$ will be called an *algebraic subset*. We remark that any ideal of $k[x_1, \dots, x_n]$ is finitely generated by Hilbert's Basis Theorem, so every algebraic subset is the zero locus of fintiely many polynomials.

Example 13.2. Let us consider some low-dimensional examples.

- The algebraic subsets of k are exactly the finite subsets and k itself.
- The algebraic subsets of k^2 are more interesting and include examples such as $V(y^2-x^3+1)$ and $V(xy)$.

Notice the following basic properties of V and I.

(i) V and I are order-reversing.
(ii) Trivially $J \subseteq I(V(J))$ and $X \subseteq V(I(X))$.
(iii) From the earlier items, $I \circ V \circ I = I$ and $V \circ I \circ V = V$. Thus, V and I induce mutually inverse bijections between the image of V and the image of I.

Theorem 13.3 (Nullstellensatz)**.** *Let $J \subseteq k[x_1, \dots, x_n]$ be an ideal. Then:*

(i) $V(J) = \emptyset$ if and only if $J = (1)$,
(ii) $I(V(J)) = \sqrt{J}$.

Proof. We see that if $J \neq (1)$, there exists a maximal ideal $\mathfrak{m} \supseteq J$. By the weak nullstellensatz there exists $a \in k^n$ such that $\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$. Therefore,

$$V(J) \supseteq V(\mathfrak{m}) = \{a\} \neq \emptyset.$$

This proves (i).

It is obvious that $\sqrt{J} \subseteq I(V(J))$. For the other direction, assume that $0 \neq f \in I(V(J))$. define

$$\tilde{J} := (J, yf-1) \subseteq k[x_1, \dots, x_n, y].$$

Clearly, $V(\tilde{J}) = \emptyset$, which by (i) implies that $(J, yf-1) = 1$. Hence there exist polynomials $p_i, q \in k[\underline{x}, y]$ and $g_i \in J$ such that

$$1 = \sum_i p_i(\underline{x}, y) \cdot g_i(\underline{x}) + q(x\underline{x}, y)(yf-1).$$

Pass to the fraction field of $k[x_1, \dots, x_n]$ and evaluate y at $\frac{1}{f}$ to obtain

$$1 = \sum_i p_i(\underline{x}, 1/f) \cdot g_i(x).$$

Multiplying by a large power of f to clear denominators we obtain the equality

$$f^N = \sum_i (f^N p_i(\underline{x}, 1/f)) \cdot g_i \in J$$

inside of $k[x_1, \dots, x_n]$. Thus, by definition, $f \in \sqrt{J}$. This finishes the proof. □

We shall call an ideal I *radical* if $\sqrt{I} = I$. We remark that:

- The radical of any ideal is radical.
- Any prime ideal is radical, but the converse is not true as $(xy) \subseteq k[x_1, \dots, x_n]$ shows.

Corollary 13.4. *The maps V and I induce inverse and order reversing bijections between the radical ideals of $k[x_1, \dots, x_n]$ and the algebraic subsets of k^n.*

Lemma 13.5. *Let R be a commutative ring and $I \subseteq R$ an ideal. Then*

$$\sqrt{I} = \bigcap_{I \subseteq \mathfrak{p} \text{ prime}} \mathfrak{p}.$$

Proof. By the correspondence theorem between ideals of R containing I and ideals of R/I, we may without loss of generality assume that $I = 0$. If $x \in \sqrt{0}$, then there exists $N \geq 1$ such that $x^N = 0$. Then, for any prime ideal $\mathfrak{p}$, $x^N = 0 \in \mathfrak{p}$ implies that $x \in \mathfrak{p}$. This shows the easy inclusion.

Suppose now that x is not in $\sqrt{0}$. It is enough to show that there is a prime $\mathfrak{p}$ such that $\mathfrak{p} \cap \{1, x, x^2, \dots\} = \emptyset$.

Let X be the collection of ideals J of R such that $J \cap \{1, x, x^2, \dots\} = \emptyset$. Then, by assumption $0 \in X$ and thus X is nonempty. Furthermore, for any chain $\{I_\alpha\}$ in X we have that $I \coloneqq \cup_\alpha I_\alpha$ is an ideal and clearly $I \in X$. This shows that all chains have an upper bound in X. By Zorn's lemma, there exists a maximal element $\mathfrak{p} \in X$. It is sufficient to prove that $\mathfrak{p}$ is prime. Suppose for the sake of contradiction that $a, b \in R$ are such that $ab \in \mathfrak{p}$ but $a \notin \mathfrak{p}$ and $b \notin \mathfrak{p}$. Then, by maximality, there exists $x^i \in \mathfrak{p} + (a)$ and $x^j \in \mathfrak{p} + (b)$ and it is easy to see that $x^{i+j} \in \mathfrak{p}$ is a contradiction. Thus, we are done. □

14. Lecture 14

The topic of this lecture is the *Zariski topology.* We will assume throughout that k is an algebraically closed field.

Lemma 14.1. *There is a topology on k^n such that the closed subsets are precisely the algebraic sets.*

Proof. We see that $k^n = V((0))$ and $\emptyset = V((1))$. Furthermore, for any collection of ideals I_α:

$$\cap_\alpha V(I_\alpha) = V(\sum_\alpha I_\alpha)$$

and

$$V(I) \cup V(J) = V(I \cap J) = V(IJ).$$

Indeed, the inclusions $IJ \subseteq I \cap J \subseteq I$ and $IJ \subseteq I \cap J \subseteq J$ imply that $V(IJ) \supseteq V(I \cap J) \supseteq V(I) \cup V(J)$ and it is therefore sufficient to show that $V(IJ) \subseteq V(I) \cup V(J)$. Assume for the sake of contradiction that there exists $a \in V(IJ)$ such that we have $f \in I$ and $g \in J$ with $f(a) \neq 0$ and $g(a) \neq 0$. This implies that fg does not vanish at a, but $fg \in IJ$, which gives a contradiction. Thus we are done. □

The above topology is called the *Zariski topology on k^n*. We remark that this topology is very coarse and is for instance not Hausdorff. In the simplest case $n = 1$ we obtain the cofinite topology on k.

Given $f \in k[x_1, \dots, x_n]$ we define the *nonvanishing set*

$$D(f) = k^n \setminus V(f).$$

Let us now give some straightforward properties of the Zariski topology.

Proposition 14.2. *Endow k^n with the Zariski topology. Then:*

(i) Points of k^n are closed.
(ii) The open subsets $D(f)$ form a basis.
(iii) k^n is compact.
(iv) k^n is a noetherian topological space, i.e. satisfies the descending chain condition on closed subsets.

Proof. For any $a \in k^n$ we have that $V(x_1 - a_1, \ldots, x_n - a_n) = \{a\}$ from which (i) follows.

If $a \in U = k^n \setminus V(I)$, then there exists $f \in I$ such that $f(a) \neq 0$. It is straightforward to see that $a \in D(f) \subseteq U$ and (ii) follows.

If $Z_\alpha = V(I_\alpha)$ is a collection of closed subsets of k^n such that $\cap_\alpha Z_\alpha = \emptyset$, then $V(\sum_\alpha I_\alpha) = \emptyset$. The Nullstellensatz implies that $\sum_\alpha I_\alpha = (1)$ and therefore there exists finitely many indices $\alpha_1, \ldots, \alpha_n$ for which $(1) = \sum_{i=1}^n I_{\alpha_i}$. We conclude that $\cap_{i=1}^n Z_{\alpha_i} = \emptyset$ and thus (iii) holds.

If $V(I_1) \supseteq V(I_2) \supseteq \cdots$ is a descending chain of closed subsets where I_k are radical, then by the Nullstellensatz $I_1 \subseteq I_2 \subseteq \cdots$ is an ascending chain of radical ideals in $k[x_1, \ldots, x_n]$. Since $k[x_1, \ldots, x_n]$ is noetherian this stabilises and (iv) follows. □

We remarkt hat any algebraic subset of k^n inherits a subspace topology from the Zariski topology on k^n.

Corollary 14.3. *If $X \subseteq k^n$ is closed, then the Zariski topology on X satisfies (i), (ii), (iii) and (iv) in the previous proposition, with $D(f) \cap X$ for (ii).*

Proof. By the correspondence theorem. □

Suppose that $V \subseteq k^n$ is an algebraic subset. Then we can define the *coordinate ring*

$$k[V] = k[x_1, \ldots, x_n]/I(V)$$

which in particular is *reduced* (has no nonzero nilpotent elements). We will call the image of the natural ring injection

$$k[V] \hookrightarrow \{\text{function } V \to k\}$$

the *regular functions* on V.

Hilbert's Nullstellensatz together with the correspondence theorem gives that there are inverse bijections V, I between

$$\{\text{radical ideals of } k[V]\} \longleftrightarrow \{\text{closed subsets of } V\}.$$

Definition 14.4. A topological space X is *irreducible* if $X \neq \emptyset$ and $X = X_1 \cup X_2$ with X_1 and X_2 closed implies $X = X_1$ or $X = X_2$.

We remark that this definition is only interesting in very coarse topologies and that it is equivalent to that any nonempty open subset is dense.

Exercise 14.5. Prove that any irreducible space is connected.

Exercise 14.6. Prove that if $Y \subseteq X$ are topological spaces and Y is irreducible, then $\bar{Y} \subseteq X$ is also irreducible.

Exercise 14.7. Prove that $V(x_1 x_2) \subseteq k[x_1, x_2]$ is not irreducible, but is connected.

Proposition 14.8. *An algebraic subset $V \subseteq k^n$ is irreducible if and only if $I(V)$ is a prime ideal if and only if $k[V]$ is an integral domain.*

Proof. If V is reducible then $V = V_1 \cup V_2$ for some closed proper subsets $V_i \subset V$. Since $I(V) = I(V_1) \cap I(V_2)$ we can choose $f_i \in I(V_i) \setminus I(V)$. Then $f_1 \cdot f_2 \in I(V)$ bur $f_i \notin I(V)$ which implies that $I(V)$ is not prime.

If $I(V)$ is not prime there exist $f_1, f_2 \in k[x_1, \ldots, x_n]$ such that $f_1 f_2 \in I(V)$ yet each $f_i \notin I(V)$. Let $V_i = V(f_i) \cap V$ which must be a proper closed subset of V since $f_i \notin I(V)$. However,

$$V = V_1 \cup V_2$$

since $f_1 f_2 \in I(V)$. □

Hence we have a correspondence between $k[V]$ and V such that

- Radical ideals correspond to closed subsets.
- Prime ideals correspond to irreducible closed subsets.
- Maximal ideals correspond to points.

Proposition 14.9. *Let X be a noetherian topological space. Then there exist irreducible closed subsets $X_i \subseteq X$ such that $X = X_1 \cup \cdots \cup X_n$ and such that $X_i \not\subseteq X_j$ for $i \neq j$. Furthermore, this decomposition is unique up to order.*

Proof. Let us first prove existence. Suppose for the sake of contradiction that X admits no such decomposition. Then in particular $X = X_1 \cup X_1'$ where X_1, X_1' are proper closed subsets and where without loss of generality X_1 does not either admit such a decomposition. Then again $X_1 = X_2 \cup X_2'$ where without loss of generality X_2 admits not decomposition and X_2, X_2' are proper closed subsets. Proceeding recursively we obtain a descending chain $X_1 \supseteq X_2 \supseteq \cdots$ which does not stabilising, contradicting the noetherian assumption.

To prove uniqueness, we claim that $\{X_1, \ldots, X_n\}$, which clearly are distinct, are exactly the maximal irreducible subsets of X. Note that if $Y \subseteq X$ is irreducible, then $Y = Y \cap X_1 \cup \cdots \cup Y \cap X_n$ and by irreducibility $Y \subseteq X_i$ for some X_i. This proves in particular that if Y is a maximal irreducible subset, then it is equal to one of the X_i. Note on the other hand that if $X_i \subseteq Y$ and Y is irreducible, then there exists j such that $X_i \subseteq Y \subseteq X_j$ and the redundancy condition implies that $i = j$ and $X_i = Y$. This finishes the proof of uniqueness up to permutation. □

Definition 14.10. The maximal irreducible subsets in a noetherian topological space are called the *irreducible components*.

Exercise 14.11. Observe that the irreducible components are always closed.